

The Role of Frequency and Optionality in the Regularization of Linguistic Variation

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Abstract

Languages are widely known to become more regular over time, yet not all variation is regularized — some forms resist regularization and remain stable over many generations. A large body of work has explored why learners regularize unpredictable variation, often focusing on the cognitive mechanisms that underlie this behavior and how these mechanisms differ between adults and children. However, less attention has been given to investigating the factors that allow some variation to remain stable over time. In this paper, we propose that optionality may be a particularly stable form of linguistic variation, likely to resist cognitive pressures to regularize. To test this hypothesis, we conducted two artificial language learning experiments designed to investigate the conditions under which optionality remains stable or becomes regularized. In Experiment 1, we compared optional variation to variation involving two alternating overt forms, expecting optionality to show greater stability. In Experiment 2, we explored whether the frequency of the optional form would influence regularization, hypothesizing that rare optional forms might resist regularization more effectively than frequent ones. Our findings suggest that, while optionality is not inherently more stable, it is nonetheless special, revealing critical insights into cognitive mechanisms proposed to be responsible for regularization behavior in child and adult learners.

1. Introduction

1.1. Regularization

Languages are constantly changing, often in ways that increase regularity over time — a process known as *regularization*. Researchers studying language change have long observed that irregular patterns tend to become regularized (Lieberman et al., 2007), and emerging signed (Senghas et al., 1997; Senghas & Coppola, 2001; Siegel, 2007) and spoken (Aitchison, 2011; Bickerton, 2020; DeGraff, 2001; Siegel, 2007) languages often evolve into more regular, structured systems. Similarly, a large body of work shows that learners exposed to inconsistent patterns tend to regularize, imposing structure and eliminating inconsistencies in their own productions. In natural language, deaf children of late-learning parents are known to regularize, allowing them to surpass their parents' inconsistent usage (Singleton & Newport, 2004). In the lab, both children and adults have been shown to regularize when learning artificial languages with unpredictable variation, with children exhibiting the strongest regularization (Austin et al., 2022; Hudson Kam & Newport, 2005, 2009), and adults showing weaker regularization that becomes amplified over generations (Real & Griffiths, 2009; K. Smith et al., 2017; K. Smith & Wonnacott, 2010). Here, regularization follows the same paths observed in natural language: learners either adopt a single dominant form (with the minority form being lost) or condition their use of the variants on some social or grammatical context (Austin et al., 2022; Hudson Kam & Newport, 2005, 2009; Schwab et al., 2018), a process sometimes referred to as *specialization*.

We know learners tend to regularize unpredictable variation, but what drives this behavior? The prevailing view is that regularization arises, at least in part, from cognitive biases that favor low entropy, more learnable systems (K. Smith et al., 2017); but the mechanisms responsible for this behavior are not fully understood. One possibility is that learners have a *determinism* bias, in which they expect variation in language to be meaningfully conditioned on something (Chomsky, 1965; Clark, 1988; Hornstein, 2009). Here, learners purportedly impose structure even when none exists in the input, thereby shifting languages toward more predictable patterns (Keogh et al., 2024). Another possibility is that *general cognitive constraints*, such as working memory limitations, drive regularization (Hudson Kam, 2019; Hudson Kam & Chang, 2009). This explanation is motivated by findings that children, who have more limited cognitive capacities, are the strongest regularizers (Austin et al., 2022; Hudson Kam & Newport, 2009). According to this view, limited working memory capacities cause learners to struggle to encode unconditioned forms during learning, fail to retrieve them during

production, or both, ultimately leading to regularization (Hudson Kam, 2019; Keogh et al., 2024; Schwab et al., 2018).

Researchers studying the cognitive reasons for regularization have historically focused on the conditions that lead learners to regularize, typically by asking whether there are circumstances that lead adults to regularize more strongly, like children do. For instance, adults tend to regularize more when told their input might contain mistakes, a way of relaxing their assumption that the input is conditioned (i.e. their ‘determinism’ bias) (Perfors, 2016). Adults also regularize more in experiments that tax working memory, either by adding many more forms to remember (Hudson Kam & Newport, 2005), or by increasing working memory load during learning or production, typically by asking learners to perform an unrelated task memory or math task (Hudson Kam, 2019; Hudson Kam & Chang, 2009; Keogh et al., 2024; Perfors, 2012).

Recently, however, researchers have begun to ask whether there are circumstances in which learners can acquire variation *without* regularizing (Hudson Kam, 2015; Samara et al., 2017; Schwab et al., 2018). This line of work is inspired by research in sociolinguistics, which finds that conditioned variation — variation that is probabilistically, not deterministically, conditioned on a context — is not only ubiquitous across languages (Aissen, 2003; D’Arcy & Tagliamonte, 2015; A. Kroch & Small, 1978; Labov, 1989), but also readily learned, not regularized, by young children (Chen & Schuler, 2024; Labov, 1989; K. Miller, 2013; J. L. Roberts, 1994; J. Smith et al., 2007). As a result, artificial language learning researchers have increasingly made a distinction between *unconditioned variation* — in which variation is completely unpredictable — and *conditioned variation* — in which variation is probabilistically conditioned on some factor internal or external to the language. When artificial languages are designed with variation that resembles sociolinguistic variation — probabilistically conditioned on particular grammatical (Hudson Kam, 2015) or social context (Samara et al., 2017; Schwab et al., 2018) — adults (and to some extent children) regularize less, though they do not always learn the precise probabilistic conditioning pattern that researchers intended.

1.2. *Stable variation*

Linguists have long recognized that not all variation leads to change. Over 50 years ago, (Weinreich et al., 1968) famously argued that “not all variability and heterogeneity in language structure involves change, but all change involves variability and heterogeneity”. Rather than

treating all variation as change in progress (Bailey, 1972), Weinrich, Labov, and Herzog (1968) observed that many cases of variation remain remarkably stable¹ over time.

One of the most well studied examples of this kind of stable variation in sociolinguistics is the alternation between *-ing* and *-in* in English (e.g. “walking” vs. “walkin’”) (Fischer, 1958; Forrest, 2015; Hazen, 2008; Labov, 1989, 2001; Tagliamonte, 2004; Tamminga, 2016; White et al., 2024). This body of work has shown that variable ING is conditioned by both grammatical and social factors: speakers are more likely to use *-in* with verbs but *-ing* with nouns, and they favor *-ing* in formal contexts while using *-in* in casual ones. Crucially, variable ING is thought to have remained stable across many generations and geographic regions despite its inherently variable use (Fischer, 1958; Labov, 1989; Labov & Labov, 2006; Trudgill, 1974). Rather than representing a gradual change in progress, ING demonstrates that some linguistic forms resist regularization and settle into a state of apparently stable variation (Labov, 1989). Of course, this is just one of many such cases: a large body of work in sociolinguistics has documented similar examples of stable variation at all linguistic levels, from phonology to syntax (Adger & Smith, 2010; Blondeau, 2001; Bouchard, 2018; Carvalho et al., 2015; Cepeda, 1995; Erker & Guy, 2012; Fox, 2006; Guy, 1980; Labov & Cohen, 1967; K. Miller, 2007; Mollica, 1991; Newkirk-Turner & Green, 2016; Orozco & Hurtado, 2020; Shin, 2016; J. Smith, 2000).

Researchers in historical linguistics have also noticed that certain linguistic properties appear to be remarkably resistant to change (Breitbarth et al., 2019). For example, Nichols (2008) in “Diversity and Stability in Language”, shows (among many other examples) that some Indo-European daughter languages preserve certain inherited features more reliably than others. Inherited gender systems are relatively stable, for example, with many languages maintaining either the original system or a simplified version of it. In contrast, noun declension classes are significantly less stable, often undergoing more substantial restructuring or simplification. Such differences suggest that certain linguistic properties, despite facing similar pressures to change, are inherently more resistant to regularization. Many other researchers have similarly demonstrated that some linguistic features remain stable across long periods, even in contexts where we might expect them to change, such as language contact and acquisition (Bickel, 1999; Field, 1998; Johanson, 2013; Maslova, 2000).

¹ To be clear, by *stable* we do not mean immutable. Following Nichols (1996), we use the term stable to describe the fact that some linguistic elements are, in her words, ‘more resistant to change, loss, or borrowing (than other elements of language). Nothing in language, of course, is truly immutable. In fact, nothing even comes close to immutability.... Linguistic stability may therefore seem to be something of a misnomer. Some elements of languages, however, are more prone to change than others, and *stable* is the best term for those that are least prone to change.’

A number of accounts have been proposed within cognitive science and linguistics to explain why some variation is regularized while other variation remains stable. One prominent hypothesis is that the conditioning itself licenses stability (Hudson Kam, 2015; K. L. Miller & Schmitt, 2012). That is, stable variation is often conditioned on some social or grammatical context (even though that conditioning is probabilistic, not deterministic), and this conditioning allows multiple variants to persist unchanged over time. Another hypothesis is that certain properties of language can counteract regularization pressure by supporting learnability of variable forms. Researchers studying complex gender marking systems, for example, have argued that these sometimes resist regularization — even though they are ‘needlessly complex’ (Lass, 1990; Trudgill, 2011; Wurzel, 1989) — when other factors facilitate their acquisition. Under this view, overly complex systems are typically regularized due to learnability constraints, but if the language provides additional structural support, the cognitive pressure to regularize is reduced. For example, while English has simplified its gender marking system over time, German has maintained a relatively complex system. Research suggests that the benefits provided by grammatical gender in discourse tracking German may outweigh the difficulty associated with acquiring these complex systems (Dahl, 2004; Watts, 2019).

We agree that both of these explanations — conditioning factors and learnability support — likely play an important role in explaining why some patterns remain stable despite learners’ cognitive biases toward regularization. In addition, we propose that *optionality* may play a role. In the following section, we argue that optionality is a particularly stable type of variation that may naturally escape the cognitive pressure to regularize.

1.3 Stability through optionality

In artificial language learning studies, researchers typically design languages with variation between two or more overt alternating forms — such as *ka* and *po* (Austin et al., 2022; Hudson Kam & Chang, 2009; Keogh et al., 2024; Perfors, 2016). In contrast, many (but certainly not all) cases of stable variation in natural language can be better characterized as optionality, where a morpheme or syntactic process is either realized or omitted. Deletion (Guy, 1980; Labov & Cohen, 1967) and lenition (Cepeda, 1995; Fox, 2006; K. Miller, 2007) are well-known examples of this, but beyond phonological processes, optionality also appears in morphosyntactic variation. For instance, in many English dialects, subject-verb agreement shows stable variation: “we was” is in stable variation with “we were” (Adger & Smith, 2010; J. Smith, 2000), which we argue can be interpreted as optional agreement. Other examples

include the variable presence of plural possessive -s and 3rd-person singular -s in English dialects (Newkirk-Turner & Green, 2016), subject pronoun expression in many Romance languages (Bouchard, 2018; Carvalho et al., 2015; Erker & Guy, 2012; Orozco & Hurtado, 2020; Shin, 2016), the alternations between *que* and *de que* in Brazilian Portuguese (Mollica, 1991) and *nous* vs. *nous autres* in French (Blondeau, 2001), and many others. While many of these patterns have traditionally been analyzed as alternations between two overt forms, we argue that they can also be understood as cases of optionality — a single morpheme or process that is either realized or omitted. .

Optionality is not just a feature of sociolinguistic variation but also emerges as a diachronically stable pattern elsewhere. One particularly well-documented example comes from a typological analysis of optional and alternating case marking (Chappell & Verstraete, 2019). Optional case marking refers to systems where a case marker may be present or absent depending on context, rather than obligatorily marking a grammatical role. Such systems are relatively common for both subjects (A) and objects (O), whereas alternating case marking — where two different case markers appear in the same context — are much rarer (Chappell & Verstraete, 2019). Optional O marking appears to be the most widespread, so much so that (Bossong, 1991) described it as “a preferred target of diachronic evolution”. Other typologically common cases of optional realization include *pro-drop* (Huang, 1984), null-subject languages (Rizzi, 1986; Valian, 1991), and optional plural marking systems (Haspelmath, 2013; Song, 1975). In these languages, subjects, pronouns, or plural marking can be optional, usually omitted when they are redundant or recoverable from discourse.

Why might optionality be particularly resistant to regularization? One possibility is that optional forms — where a single morpheme or process is either realized or omitted — may escape the principle of contrast (Clark, 1988) or mutual exclusivity (Markman & Wachtel, 1988), which bias learners toward mapping distinct forms to distinct meanings. Relatedly, optionality may naturally afford learners a way to condition along a continuous dimension rather than a categorical one. Fruehwald and Wallenberg (2013) have argued that, even in cases of two overt forms, specialization — regularization in which each form is conditioned on a particular context — is only possible when the dimension of conditioning is categorical. If the context is continuous (or ordinal), it may be impossible for forms to fully specialize. Optional forms may naturally accommodate continuous conditioning by allowing learners to optionally realize the form along a gradient, reflecting degrees of properties (e.g. emphasis, formality, discourse salience, etc) rather than encoding two categorically distinct meanings. Another possibility has to do with communicative efficiency. Optionality may reflect a cognitively “safe” (even preferred) form of

innovation, allowing learners to be variable by reducing redundancy (i.e. omitting elements that are recoverable from other aspects of the structure or discourse), without the memory burden of an additional form-meaning mapping.

1.4 The current study

The observation that many cases of stable variation can be interpreted as optionality raises the question: is there something special about optional forms that makes them inherently more stable? Although optional variation is widely attested across languages, little direct empirical evidence exists about how learners handle optionality when faced with cognitive pressures to regularize. In the present study, we provide a first test of this hypothesis through two artificial language learning experiments, designed specifically to isolate optionality as a potential stabilizing factor. In Experiment 1, we exposed participants to a plural marker that varied unpredictably — the type of variation known to be especially vulnerable to regularization — and asked whether an optional marker (ka) would be less likely to be regularized than two alternating forms (ka and po). In Experiment 2, we explored whether the frequency of an optional form influences its susceptibility to regularization, specifically testing if rare optional forms resist regularization more effectively than frequent ones. By directly examining optionality in controlled settings, these experiments offer new insights into the cognitive underpinnings of language stability.

2. Experiment 1

Our goal in Experiment 1 was to investigate whether an optional marking system would be more stable (i.e. less likely to be regularized) than an alternating marking system. Based on evidence from natural language variation and change, we proposed that optionality plays an important role in explaining why some patterns remain stable, despite learners' cognitive biases to regularize (in addition to other known factors like conditioning context and learnability constraints). We hypothesized that an optional marking system would be inherently more stable, even in circumstances under which learners normally regularize (i.e. when variation is highly unpredictable, and not conditioned on any grammatical or social context). To test this hypothesis, we exposed participants to an artificial language in which plural marking varied unpredictably, either as one optional form (*Optional* condition) or two alternating forms (*Alternating* condition). Because the singular form was unmarked ($-\emptyset$) in both of these

conditions, we included a control condition (*Alternating same-as-singular* condition) in which the singular form was marked with one of the markers from the alternating condition (e.g. singular was marked -po and alternating plural markers were -ka and -po). To determine if participants regularized, on each test trial, we showed them a plural image and asked which of two plural sentences in the artificial language was best.

We predicted that participants would regularize significantly less in the *Optional* language, where one plural marker was optionally realized, compared to the languages in which the variable plural markers were two distinct alternating forms. Further, we predicted that this should be the case, even when the singular form was overly marked with one of the variable plural markers. That is, alternating forms should be more likely to regularize, regardless of whether one of the alternating forms is homophonous with the singular form.

2.1 Methods

2.1.1 Participants

Participants were 61 adults recruited and run on Prolific Academic (www.prolific.ac). We recruited only monolingual native English speakers between 18-46 years of age who had at least an 85% study completion rate on Prolific. Participants were divided into three plural marking conditions: *Optional* (n=20), *Alternating* (n=21), and *Alternating same as singular* (n=20). All participants received \$9/hour for their participation.

2.1.2 Materials and procedure

During the exposure phase, we exposed participants to an artificial language with either optional or alternating plural markers. On each of 72 exposure trials, participants saw a picture of either one (singular) or more than one (plural) object and heard a sentence describing the picture in the artificial language. There were six possible objects in the experiment (*pig, cow, sheep, tractor, truck, and wheelbarrow*), which we randomly assigned to six novel nouns for each participant (*mawg, geed, spad, daffin, flugat, and clidam*). Each noun was presented 3 times in its singular form (paired with one object) and 9 times in its plural form (paired with more than one object). All sentences started with *gentif*, a verb meaning “there is/are”, followed by the noun. Singular and plural sentences were either marked or unmarked according to the language condition (described in section 2.1.3 below). Of the 9 plural trials for each noun, 6

were marked with the dominant plural marker (*-ka*) and 3 were marked with the minority plural marker (see Table 1).

Following the exposure phase, we gave participants a two-alternative forced-choice test. On each test trial, participants saw a picture of more than one object (plural) and were asked which of the two possible plural sentences was best. There were 36 total test trials, 6 for each of the 6 objects. We randomized the order of these test trials for each participant. Sample exposure and test trials are shown in Figure 1.

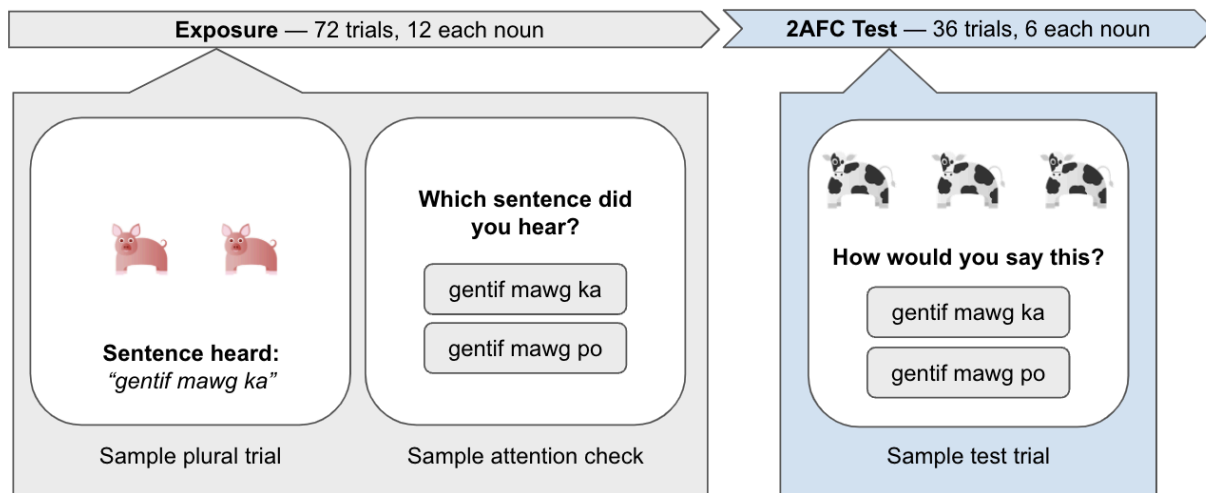


Figure 1. Schematic of study procedure including sample trials for the exposure and test phases.

2.1.3 Conditions

We assigned participants to one of three plural marking conditions: *Alternating* ($n=21$), *Optional* ($n=20$), or *Alternating same as singular* ($n=20$). In the *Optional* condition, singular sentences were unmarked ($-\emptyset$) and plural sentences were either marked with *-ka* or unmarked ($-\emptyset$). In the *Alternating* condition, singular sentences were unmarked and plural sentences were marked with one of two alternating forms: *-ka* or *-po*. To control for the fact that the unmarked plural form in the *Optional* condition was the same as the singular form, we created a third condition — *Alternating same as singular (sas)* — in which plural sentences were marked the same way as in the *Alternating* condition, but singular sentences were marked with *-po*, one of the alternating plural forms. In all conditions, plural sentences were more frequently marked with *-ka* (the dominant marker), but marker use was not conditioned by any other context. That is,

whether a plural sentence would be marked or unmarked could not be predicted by noun, animacy, or any other conditioning context. We describe these conditions in Table 1, including the proportion of use of each marker in singular and plural contexts.

Table 1. Singular and plural markers for all experimental conditions

Experiment	Condition	Singular	Plural	
			67%	33%
1	Optional	-∅	-ka	-∅
	Alternating	-∅	-ka	-po
	Alternating (sas)	-po	-ka	-po
2	Optional frequent	-∅	-ka	-∅
	Optional rare	-∅	-∅	-ka

2.1.4 Analysis approach

We conducted our analysis in R (R Core Team, 2020) via Google Colab, a cloud-based Jupyter notebook (Kluyver et al., 2016). Figures and general data wrangling were accomplished with the *tidyverse* package (Wickham et al., 2019).

To determine the effect of condition on participants' use of the dominant plural marker (-ka), we built a logistic mixed-effects regression model predicting use of the -ka marker by condition (*lme4* package: Bates et al., 2015; *lmerTest* package: Kuznetsova, Brockhoff & Christensen, 2017). We included condition as a simple-coded fixed effect and by-participant and by-object random intercepts in the model formula:

$$(1) \text{ isKa} \sim \text{condition} + (1|\text{participant}) + (1|\text{object})$$

To further determine whether participants made the language more regular, we used entropy of participant responses to quantify regularization of the dominant form, and mutual information to quantify whether participants conditioned their marker choice on the noun or the animacy of the noun. Both entropy and mutual information were calculated with the *infotheo* package (Meyer, 2014). To determine whether either of these regularization metrics differed by condition, we built two additional regression models to predict participant entropy or mutual information by condition. For the entropy model, we used simple linear regression to predict entropy by condition, since each participant has only one value capturing the entropy of their test-phase responses. For the mutual information model, we used mixed-effects regression to predict mutual information by condition, conditioning context (noun or animacy), and their

interaction. We included by-participant random intercepts in the mutual information model formula because each participant has two values for mutual information (one for each conditioning context):

$$(2) \text{ mutual information} \sim \text{condition} * \text{context} + (1|\text{participant})$$

In all models, we determined the significance of main effects and interactions using model comparison (Barr et al., 2013, Bates et al., 2015).

2.2 Results

2.2.1 Attention check

To ensure participants were listening carefully during exposure, after each exposure trial we asked them to select (from two written options) which sentence they just heard. We analyzed the results from this attention check first, intending to use poor performance (<85% accuracy) as a basis for exclusion. However, as shown in Table 2, mean accuracy was near the ceiling (99%) in all conditions and no participants were excluded for failing to pay attention during exposure.

Table 2. Exposure phase accuracy during Experiment 1

	Mean	SD	Min	Max
Optional	99.38	0.71	98.61	100
Alternating	99.40	0.70	98.61	100
Alternating (sas)	99.17	2.04	91.67	100

2.2.2 Use of the -ka marker

We hypothesized that a plural marking system with one optional form would be more stable than a plural marking system with two alternating forms, predicting that participants exposed to alternating markers would regularize more than those exposed to an optional marker. As a first test of this hypothesis, we asked whether participants in the *Alternating* conditions use the dominant form more than participants in the *Optional* condition. We found that, contrary to our prediction, participants exposed to two alternating plural markers used the dominant form significantly less than those exposed to an optional marker (see Figure 2). Specifically, our analysis revealed a significant main effect of condition, where participants in

both alternating conditions had significantly lower odds of selecting the marked form at test ($\chi^2(2)=20.56$, $p<0.001$; *Alternating*: $\beta=-2.49$, $SE=0.57$; *Alternating (sas)*: $\beta=-2.26$, $SE=0.57$). As seen in Figure 2, over half of all participants in the *Optional* condition made the language fully regular, using the optional *-ka* marker on every test trial.

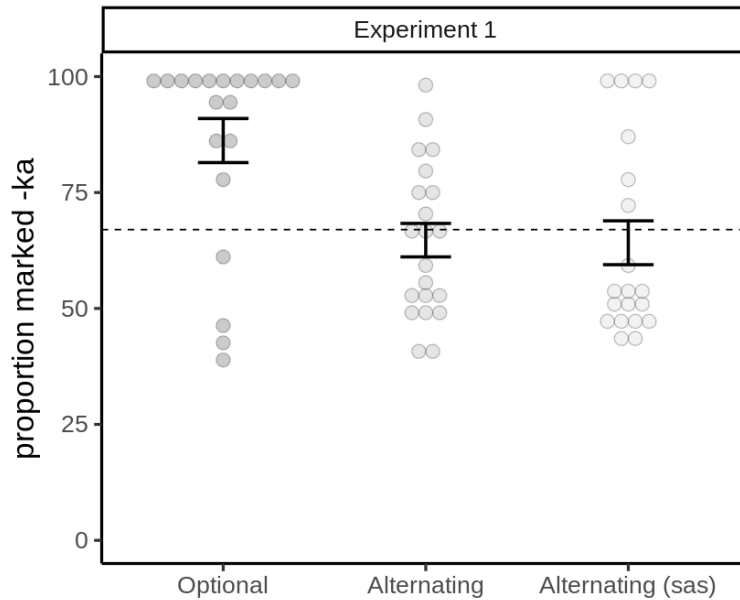


Figure 2. Proportion of trials marked *-ka* by condition. Each dot is a participant, error bars are standard error, and the dashed line is the proportion of plural sentences marked *-ka* in the input.

2.2.3 Entropy and mutual information

Evaluating use of the dominant form tells us whether participants' use of *-ka* differed significantly by condition, but not necessarily whether they regularized the language. Importantly, learners could regularize not only by maximizing the regular form, but also by maximizing the minority form (the other plural marker), or by conditioning their input in some way — for example, by lexical item or object animacy. To test this, we analyzed the entropy and mutual information of participant responses.

Entropy provides a measure of how regular a participant's responses are overall. In our artificial language, an entropy of 0 would indicate that the participant always used one marker (completely regular) and an entropy of 1 would indicate that the participant used both markers equally often. As shown in Figure 3, our entropy analysis revealed a significant main effect of condition. Entropy was significantly higher — indicating responses were significantly *less* regular — for participants in the alternating conditions ($F(2)=13.05$, $p<0.001$; *Alternating*: $\beta=0.35$,

SE=0.07; *Alternating (sas)*: $\eta^2=0.30$, SE=0.07). Again, contrary to our prediction, participants in the *Optional* condition regularized the most, with many participants making the language fully regular.

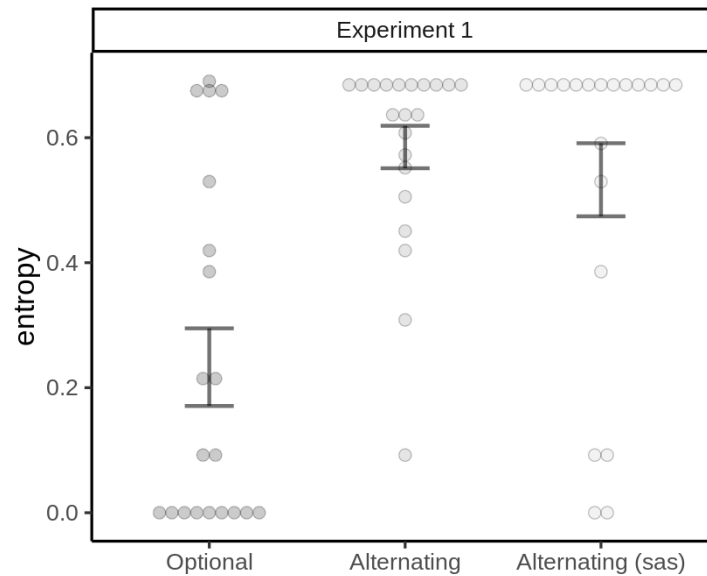


Figure 3. Entropy of plural marking by condition. Each dot is a participant and error bars are standard error. Lower entropy indicates more regularization.

To find out whether participants regularized by conditioning their marker use on a particular context (e.g. noun or animacy), we also calculated the mutual information for both of these conditioning contexts for each participant. Mutual information quantifies the amount of information gained about one random variable (here, the marker the participant used) by observing another (here, the noun or the animacy of the noun). For mutual information, higher values correspond to more regularization, indicating participants are conditioning their marker choice on that particular context to some quantifiable extent. If no information is gained by knowing the conditioning context (noun or animacy), mutual information will be 0.

We found that both condition ($\chi^2(2)=6.21$, $p=0.04$) and conditioning context ($\chi^2(1)=28.82$, $p<0.001$), but not their interaction ($\chi^2(2)=1.73$, $p=0.42$), were significant predictors of mutual information. These results reflect what we see in Figure 4: participants were indeed more likely to condition marker use in the alternating conditions, and more likely to condition marker use on the noun, compared to animacy (see Table 3 for a summary of coefficients in all models). However, the individual data points in Figure 4 reveal that mutual information was in general quite low — for many participants, at or near 0 for both conditioning

contexts — indicating most participants did not condition their marker choice on either the noun nor animacy in any condition.

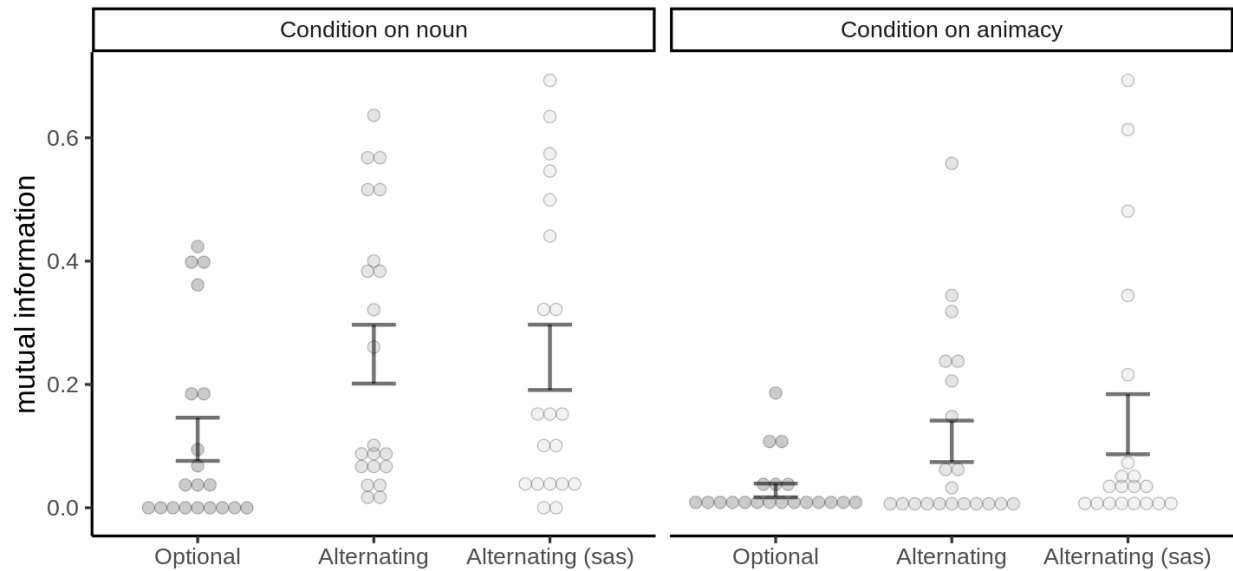


Figure 4. Mutual information by condition and context (noun or animacy). Each dot is a participant and error bars are standard error. Higher mutual information indicates that plural marking is more conditioned on that context (noun or animacy).

Table 3. Model coefficients for regression models in Experiment 1.

	Estimate	SE	z	p
Use of -ka marker				
(Intercept)	1.71	0.34	5.03	<0.001
Alternating	-2.49	0.57	-4.38	<0.001
Alternating (sas)	-2.26	0.57	-3.93	<0.001
	Estimate	SE	t	p
Entropy				
(Intercept)	0.45	0.03	14.85	<0.001
Alternating	0.35	0.07	4.76	<0.001
Alternating (sas)	0.30	0.07	4.00	<0.001
Mutual Information				
(Intercept)	0.09	0.02	3.85	<0.001
Alternating	0.08	0.06	1.39	0.10
Alternating (sas)	0.11	0.07	1.85	0.06

Noun	0.11	0.02	5.97	<0.001
Alternating:Noun	0.06	0.05	1.29	0.20
Alternating (sas):Noun	0.03	0.05	0.56	0.50

2.3 Discussion

In Experiment 1, we used an artificial language learning experiment to investigate whether an optional plural marker would be inherently more stable than two alternating forms. We predicted that participants exposed to the optional language would regularize significantly less than those in the alternating condition. However, to our surprise, we found the opposite pattern. Participants exposed to the optional language were significantly *more* likely to regularize than those in the alternating conditions. So much so that over half of the participants in the *Optional* condition regularized fully, eliminating the variation completely by using the optional marker as if it were obligatory in plural contexts. This result is striking because adults typically do not regularize this strongly in artificial language learning experiments (Austin et al., 2022; Hudson Kam & Newport, 2005, 2009), even in cases where researchers are pushing them to regularize as strongly as possible (e.g. via working memory manipulations, etc) (Hudson Kam & Chang, 2009; Keogh et al., 2024; Perfors, 2016). We return to this surprising result in the general discussion.

Why do our participants show the opposite pattern that exists in natural language typology? Recall that, in natural language, optional marking systems are more stable, while alternating systems are more prone to regularization. Yet, in our artificial language, the alternating marking systems remained fairly stable while the optional one was regularized. One possibility is that optionality resists cognitive pressures to regularize only when it is reinforced by other known stabilizing forces, such as conditioning contexts or learnability. We agree this is likely part of the explanation. However, another possibility is that the frequency of the optional form plays a role. Across languages, persistent variable forms often appear more rarely in their optional contexts, and rates of language change tend to slow down as forms become sparser (e.g. Song 1975). Perhaps optionality only can lead to stable variation, even without additional stabilizing factors, but only when the optional form itself is rare. This is the hypothesis we investigate next in Experiment 2.

3. Experiment 2

In many cases of stable variation, optional forms are more likely to be omitted than realized. One particularly well-studied example comes from French negation, where *ne* is obligatory in written language but rarely realized in colloquial speech (Armstrong, 2002; Armstrong & Smith, 2002; Ashby, 1981, 2001). Despite its extreme infrequency in French children’s input (Choi, 1986; Culbertson, 2010; Sankoff & Vincent, 1980), French children have been shown to acquire *ne* and the constraints on its use from a very early age (Chen & Schuler, 2024; Palasis, 2015). Similarly, in many pro-drop languages like Spanish and Portuguese, pronouns are frequently omitted in conversational speech, and are therefore only rarely realized in children’s language input. Yet, even with their low frequency, children acquiring pro-drop languages have no trouble learning these pronouns and mastering the constraints on when to realize them (Shin, 2016). Taken together, these patterns suggest that *infrequent* optional forms — optional forms that are rarely realized — may be particularly stable over time owing in part to being particularly learnable by children. In fact, some researchers have argued that rates of language change tend to slow down as optional forms become rarer (Bailey, 1972; A. S. Kroch, 1989), further contributing to their stability.

In Experiment 2, we tested whether the frequency of an optional form influences its stability. Specifically, we asked whether an optional plural marker would be less likely to be regularized if it was realized only rarely in the input. Participants were exposed to the same artificial language as in Experiment 1. One condition — *Optional frequent* — was identical to the *Optional* condition, where the optional plural marker was realized 67% of the time. To test the role of frequency, we introduced a second condition, *Optional rare*, which was identical in all ways except that the optional plural marker was realized much more rarely (only 33% of the time).

As in Experiment 1, to determine whether participants would regularize, we asked participants to choose which of two sentences best describes a plural image. Based on prior research suggesting that rarely realized optional forms are particularly stable, we predicted that, while our *Optional frequent* language should be regularized just as in Experiment 1, the *Optional rare* language should resist regularization and remain stable.

3.1 Method

3.1.1 Participants

We recruited and ran another 40 adults on Prolific Academic (www.prolific.ac) for Experiment 2. As in Experiment 1, all participants were monolingual native English speakers between 18-46 years of age who had at least an 85% completion rate on Prolific. Participants were divided into two plural marking conditions: *Optional frequent* (n=20) — a replication of the *Optional* condition of Experiment 1 — or *Optional rare* (n=20). All participants were paid \$9/hour for their participation.

3.1.2 Materials and procedure

The procedure was as in Experiment 1 (section 2.1.2), except for the particular plural marking conditions participants were exposed to. We describe these conditions in the following section.

3.1.3 Conditions

In Experiment 2, we assigned participants to either the *Optional frequent* (n=20) or the *Optional rare* (n=20) plural marking condition. The *Optional frequent* condition was a replication of the *Optional* condition in Experiment 1 with a new sample of participants and a more precise name. Recall that singular sentences in this condition were unmarked (-∅) and plural sentences were optionally marked with *-ka*. We refer to this condition as *Optional frequent* in Experiment 2 because the optional plural form occurs frequently in exposure (67% of plural trials were marked). The *Optional rare* condition was the same as the *Optional frequent* condition, except the optional plural form occurred more rarely during exposure: just 33% of plural trials were marked with *-ka*. As in Experiment 1, whether a plural sentence would be marked or unmarked could not be predicted by noun, animacy, or any other conditioning context. Refer to Table 1 for the distribution of markers and their proportion of use in singular and plural contexts for each condition.

3.1.4 Analysis approach

As in Experiment 1 (section 2.1.4).

3.2 Results

3.2.1. Attention check

As in Experiment 1, we first analyzed the results from the attention check in which we asked participants to select which sentence they heard (from two written options) after every exposure trial. One participant failed to achieve our 85% threshold on this task (scoring 50%) and was excluded from further analysis. For the remaining participants, mean accuracy was near the ceiling in all conditions.

Table 4. Exposure phase accuracy during Experiment 2 for included participants

	n	Mean	SD	Min	Max
Optional frequent	20	97.92	2.80	88.89	100
Optional rare	19	99.05	1.54	95.83	100

3.2.2. Use of the -ka marker

We hypothesized that an optional plural marking system with a rare optional form would be more stable than one with a frequent optional form. Specifically, we predicted that participants in the *Optional rare* condition would regularize significantly less than those in the *Optional frequent* condition. To test this prediction, we first asked whether participants in the *Optional rare* condition used the optional form less than participants in the *Optional frequent* condition. In line with our prediction, we found that participants exposed to a rarely occurring optional plural marker (-ka) used the marker significantly less than those exposed to the same optional plural marker occurring frequently (see Figure 5). Specifically, our analysis revealed a significant main effect of condition, where participants in the *Optional rare* condition had significantly lower odds of selecting the marked form at test ($\chi^2(1)=12.32$, $p<0.001$; *Optional rare*: $\beta=-2.80$, $SE=0.75$).

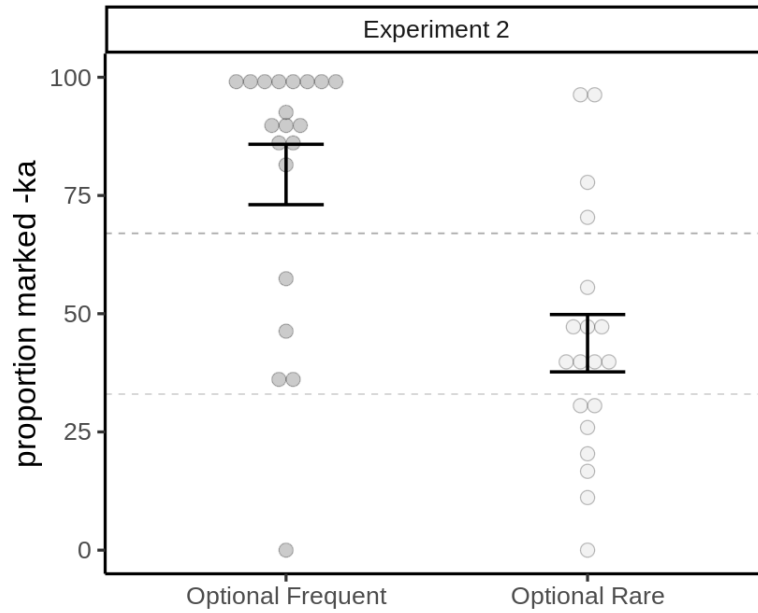


Figure 5. Proportion of trials marked *-ka* by condition. Each dot is a participant, error bars are standard error, and the dashed lines are the proportion of plural sentences marked *-ka* in the input (67% in Optional Frequent and 33% in Optional Rare)..

3.2.3. Entropy and mutual information

As in Experiment 1, we also analyzed the entropy to determine whether participant responses were regular in another way, beyond simply maximizing use of the overt plural marker at test. Recall that entropy provides a measure of how regular participant responses are overall, with an entropy of 0 indicating the participant fully regularized the language (using the same marker, either *-ka* or $\emptyset$, on all test trials). Our analysis revealed a significant main effect of condition ($F(1)=11.37$, $p=0.002$), indicating that responses from participants exposed to the *Optional rare* language had significantly higher entropy ($\eta^2=0.25$, $SE=0.08$, $t=3.37$, $p=0.002$). Thus, as predicted, participants exposed to the language with a rarely occurring optional plural marker regularized the language significantly less (see Figure 6).

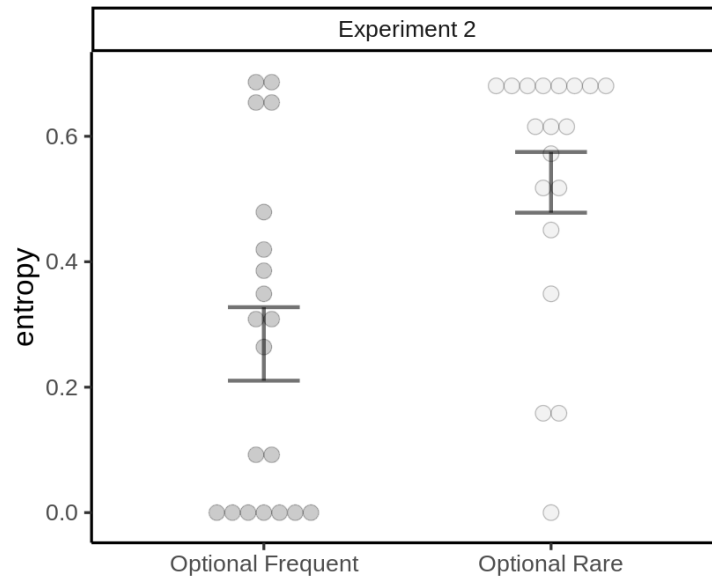


Figure 6. Entropy of plural marking by condition. Each dot is a participant and error bars are standard error. Lower entropy indicates more regularization.

Finally, we calculated the mutual information in participant responses, as in Experiment 1, to determine whether participants were conditioning their marker use in some way. Recall that mutual information provides a measure of the information gained about one random variable (marker choice) by observing another (noun or animacy), with larger values indicating participants have conditioned on that particular context to a greater extent. We found that conditioning context ($\chi^2(1)=20.11$, $p<0.001$), but not condition ($\chi^2(1)=1.30$, $p=0.25$) or their interaction ($\chi^2(1)=0.61$, $p=0.43$), were significant predictors of mutual information. These results reflect what we see in Figure 7: participants were indeed more likely to condition marker use on the noun. However, overall mutual information for both conditioning contexts was at or near zero for most participants, indicating that marker choice was conditioned on context only to a limited extent (see Table 5 for all model coefficients).

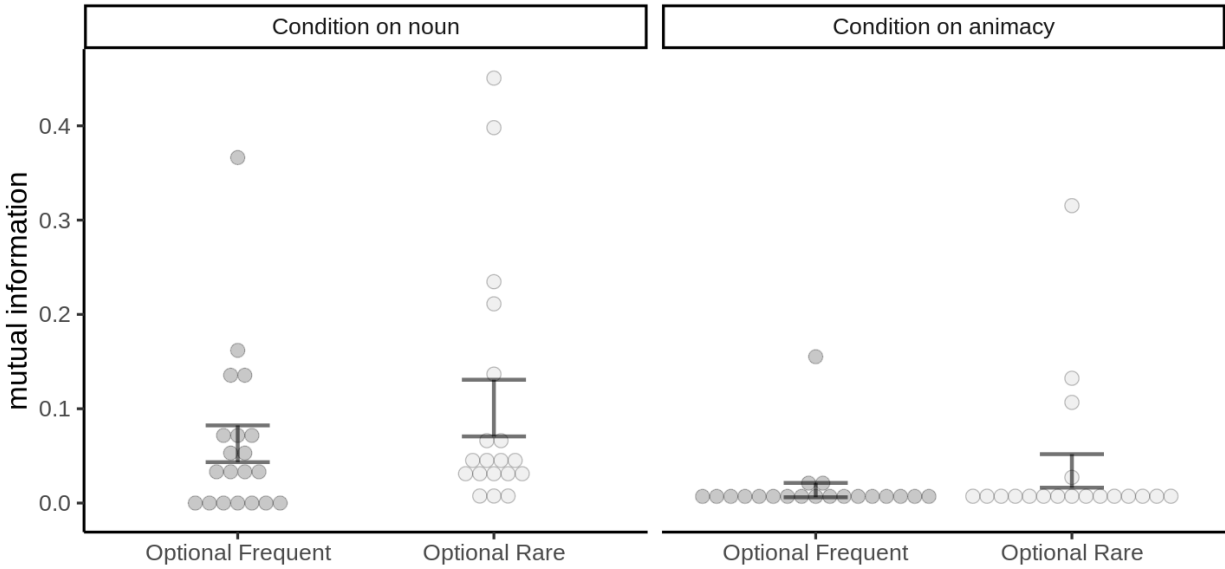


Figure 7. Mutual information by condition and context (noun or animacy). Each dot is a participant and error bars are standard error. High mutual information indicates that participant plural marking is more conditioned on that context (noun or animacy).

Table 5. Model coefficients for regression models in Experiment 2.

	Estimate	SE	z	p
Use of -ka marker				
(Intercept)	1.08	0.38	2.85	0.004
Optional rare	-2.80	0.76	-3.75	<0.001
	Estimate	SE	t	p
Entropy				
(Intercept)	0.40	0.04	10.41	<0.001
Optional rare	0.26	0.08	3.37	0.002
Mutual Information, Noun				
(Intercept)	0.02	0.01	1.68	0.10
Optional rare	0.02	0.03	0.72	0.48
Noun	0.06	0.01	5.05	<0.001
Optional rare:Noun	0.02	0.02	0.76	0.45

3.3 Discussion

Experiment 2 provides support for the idea that, in the absence of other stabilizing forces, rarely realized optional forms resist regularization. Participants in the *Optional rare*

condition, where the plural marker was present only 33% of the time, were significantly less likely to regularize than those in the *Optional frequent* condition. In fact, their behavior closely resembled the Alternating conditions from Experiment 1, where participants showed behavior closer to probability matching rather than regularization.

At the same time, just as in Experiment 1, the strong tendency toward regularization in the *Optional frequent* condition is striking, given prior evidence that adults do not typically regularize to this degree. This suggests that optionality alone may not be inherently more stable, but that optionality combined with another stabilizing force — such as low frequency — may be crucial. The interaction between optionality and frequency raises important questions about how properties of the input jointly influence stability (or regularization). We return to these issues in depth in the general discussion.

4. General discussion

In this paper, we join many others in arguing that not all variation gets regularized (Hudson Kam, 2015; Samara et al., 2017). While most research has focused on the cognitive mechanisms that drive regularization, we argue that investigating the factors that contribute to stability is equally important ((Labov, 1989; Weinreich et al., 1968). As a step in this direction, we proposed that *optionality* — where a single overt form is sometimes realized and sometimes omitted — may be a form of variation that is especially resistant to regularization. To test this hypothesis, we conducted two artificial language learning experiments. In Experiment 1, we asked whether an optional plural marker (ka) would be more stable than two alternating plural forms (ka and po). In Experiment 2, we tested whether the frequency of an optional form affects how likely it is to be regularized, specifically asking whether rare optional forms resist regularization more than frequent ones.

In Experiment 1, we expected participants in the optional condition to regularize less, but instead, they did the opposite — regularizing far more strongly than those in the alternating condition. In fact, over half of the participants eliminated variation entirely, treating the optional marker as obligatory in plural contexts. This is striking because adults typically do not regularize this strongly in artificial language learning experiments, even under conditions designed to encourage regularization (Hudson Kam 2019; Hudson Kam & Chang 2009; Keogh et al. 2024; Perfors, 2012, 2016). Moreover, this result seems to contradict patterns in natural language, where optional marking systems tend to persist while alternating systems are more prone to regularization (e.g. Chappell and Verstrate, 2019).

One explanation for this contradiction could be that optionality alone is not enough to promote stability — it may need to interact with other stabilizing forces in order to escape cognitive pressure to regularize. We proposed that frequency could be one such support, given evidence that rare forms not only resist change but also are acquired by children (Chen & Schuler, 2024). Experiment 2 tested this hypothesis by manipulating how often the optional plural marker appeared. Participants in the optional rare condition (where the plural marker appeared only 33% of the time) were indeed much less likely to regularize than those in the optional frequent condition (which was again strongly regularized). In fact, their behavior looked more like probability matching, similar to participants in the alternating conditions from Experiment 1 and many previous studies with adult learners (e.g. Austin et al, 2022; Hudson Kam, 2015; Hudson Kam & Newport, 2005, 2009; Samara et al, 2017).

4.1. Are optional forms inherently more stable?

Taken together, our results suggest that optional forms are not inherently more stable — if they were, we would expect to see stability in both of our optional conditions. Instead, our optional plural marker was strongly regularized when it was realized frequently (67%), only escaping this pressure (i.e., remaining stable) when it was realized rarely (33%). Still, optionality patterned differently from other types of variation. Across all conditions, adults generally responded as predicted by prior work (Austin et al., 2022; Hudson Kam & Newport, 2005, 2009): when exposed to unpredictable variation with no external pressure to regularize (e.g., no working memory demands), they preserved the input and maintained variability. The one exception was the optional frequent condition, where our adult participants' behavior was dramatically different — rather than maintaining variation, most learners eliminated it entirely, making the language fully regular and deterministic.

At first glance, this raises the possibility that it is *rareness*, not optionality, that protects forms from being lost. Recall that both of our alternating conditions also included a rare form (po), and these conditions remained similarly stable. Indeed, this distribution — in which one form is frequent (dominant) and another is rare (minority) — is common in studies of regularization. This account aligns with broader findings in linguistics suggesting that language change slows down as forms become sparse (Bailey, 1972; A. S. Kroch, 1989). When a form is rarely attested, it may stand out perceptually, increasing its salience to learners and making it less likely to be eliminated through regularization. From a cognitive perspective, this could mean

that low-frequency forms are more easily noticed and retained, even in cases where regularization might otherwise be expected.

The idea that rareness itself might protect forms from being lost explains why our optional frequent condition — the only condition lacking a truly rare form — was uniquely vulnerable to regularization. However, this interpretation assumes that a null form is essentially "nothing," a strong assumption given that null forms can represent many things in natural languages. If we do treat the null as merely absence, we face another challenge: what mechanism prevents the overt form from regularizing in the optional rare condition? Furthermore, this account does not straightforwardly explain children's behavior. Children robustly regularize even in conditions involving two overt forms, continuing to do so even when the dominant form becomes relatively rare (as low as 40%; (Austin et al., 2022). At the same time, children do apparently acquire sociolinguistic variation that resembles optionality. (Chen & Schuler, 2024; Labov, 1989; K. Miller, 2013; J. Roberts, 1997; Shin, 2016). In the following sections, we address each of these in turn, first by considering specific cognitive mechanisms that might drive regularization, and then by examining how these mechanisms differ between adults and children.

4.2. The cognitive mechanisms for regularization

A large body of work has explored the cognitive mechanisms behind regularization, often by trying to push adults to behave more like children — regularizing more than expected — to uncover the underlying causes of this behavior. Many studies have succeeded in amplifying regularization in adults, but never to the extent seen in children. In fact, some have argued that adult regularization is merely a weaker version of the same process, which only strengthens over generations (Reali & Griffiths, 2009; K. Smith et al., 2017; K. Smith & Wonnacott, 2010). This makes our findings particularly striking: in the current experiments, adults in our optional frequent conditions regularized as strongly as children typically do. Understanding what drove this effect could offer a new window into the cognitive mechanisms behind regularization. Do existing accounts — e.g., a bias for determinism, or working memory limitations — help explain why adults regularize so strongly in our optional frequent language?

Turning first to the determinism bias, recall that learners have been argued to expect variation in language to be meaningfully conditioned (Clark, 1988; Chomsky, 1965; Hornstein, 2009), leading them to impose structure even when none exists in the input (Keogh et al., 2024; Samara et al., 2017; Smith & Wonnacott, 2010). Indeed, this tendency extends beyond

language: humans frequently misinterpret random or unstructured sequences by attributing meaningful patterns to them, even when such structure is absent (Bar-Hillel & Wagenaar, 1991; Wolford et al., 2004). There are two plausible interpretations of our results in light of this bias. One possibility is that the determinism bias proceeds unchecked when the input clearly affords a pattern that reduces uncertainty. In our optional frequent condition, the singular-plural distinction was highly consistent — plural nouns reliably took *ka*, while singular nouns were unmarked — with only a few exceptions. Additionally, this distinction aligns with the singular-unmarked/plural-marked pattern already present in English, potentially making the rule even more salient or accessible to our participants. Under these conditions, adults may readily detect and reinforce this rule, leading to strong regularization. This account could also explain why a small number of adults in the *alternating same as singular* condition appeared to regularize as well (though most did not): even there, a mostly consistent singular-po/plural-*ka* rule was available. Alternatively, along similar lines, adults might interpret infrequent omissions of the optional form as errors — small deviations from an otherwise consistent pattern — and thus regularize by eliminating these perceived mistakes (Perfors, 2016).

Another common account emphasizes working memory limitations, either during encoding (where learners fail to represent variable input accurately) or during production (where retrieval failure leads to regularization). However, it is difficult to see how working memory limitations alone could fully explain our pattern of results. One possibility is that adults regularize in the optional frequent condition due to encoding failures. However, for encoding failure to specifically target the optional frequent condition — and not the optional rare condition — we would need an additional explanatory mechanism, such as the determinism bias described earlier. Alternatively, regularization might arise due to retrieval and production difficulties. Recent accounts propose that regularization occurs during production, driven by retrieval-based mechanisms similar to priming, whereby previously produced forms reinforce subsequent selections (Goldberg & Ferreira, 2022; Hudson Kam & Chang, 2009; Keogh et al., 2024; Schwab et al., 2018). Under this explanation, our optional frequent condition might promote regularization because null forms are inherently weaker and thus less reinforcing compared to overt forms. This retrieval-based account could also help explain why, in our optional rare condition, many participants appeared to overshoot the input distribution — perhaps overweighting the overt form due to its greater salience or retrievability, even though it was less frequent than the null form.

Resolving this issue may require a deeper consideration of the nature of the null form itself. Null forms can be analyzed in multiple ways — they could be underlying morphemes with

no overt realization, phonologically reduced forms, or even a complete absence of marking. Crucially, how learners represent nulls might influence their likelihood of regularization. If null forms simply “do not exist” in learners' mental representations, then it is perhaps unsurprising that learners regularized when the optional form was frequent. But how, then, can we explain the stability observed when the optional form was rare? Could this stability suggest that learners sometimes flexibly reinterpret null forms as genuine morphological variants (i.e., null allomorphs), depending on the structure and frequency of the input? Understanding how learners represent and interpret null marking might thus be critical for explaining why optionality sometimes resists regularization. More broadly, understanding regularization itself may require a deeper exploration of how learners interpret the presence — and especially the absence — of forms in their input.

4.3 Regularization in children

Our findings have important implications for understanding regularization in children. Although children's strong regularization tendencies are often explained by working memory limitations, our results suggest that additional factors are at play.

The “Less is More” hypothesis is often used to explain why children regularize more strongly than adults, attributing this tendency to children's more limited cognitive capacities (Newport, 1990). This limitation is usually interpreted as differences in working memory between children and adults; however, others have suggested children may instead structure linguistic input differently from adults (Austin et al., 2022; Newport, 1990). For example, adults might store linguistic forms lexically (e.g., cows take ka), while children might generalize more categorically (e.g., nouns take ka), or vice versa (e.g. Austin et al., 2022). If so, adults in our study may have behaved more like children not due to memory limitations per se, but because the structure of our input encouraged a more categorical representation. Indeed, our findings show adults regularizing more like children than previously observed, suggesting that factors beyond working memory alone may drive regularization. Future research should further explore these alternative explanations.

Relatedly, no single cognitive mechanism fully explains our observed patterns of behavior. Instead, regularization likely results from an interplay of multiple biases and mechanisms that learners flexibly employ. Part of the difference we observe between regularization behavior in children and adults may stem from differences in this flexibility. For example, differences in inhibitory control could allow adults to dynamically adjust their behavior

by integrating multiple cues, whereas children may struggle with such flexibility (Novick et al., 2005; Thompson-Schill et al., 2009). Adults might override their regularization biases when motivated to avoid ambiguity or uncertainty, whereas children may more rigidly adhere to forms generated by their internalized grammar. Furthermore, the structure of the input itself could trigger shifts in interpretation — for instance, allowing them to interpret null forms flexibly as either an absence of marking or as a null allomorph, depending on context.

Finally, it remains an empirical question what specific role optionality plays in children's regularization behavior. Considering the prevalence of optionality in stable linguistic variables — the kind of variation children are known to learn rather than regularize — it seems likely that optionality significantly influences children's regularization tendencies. While the optional frequent condition was a special case for adults — prompting unusually strong regularization — the optional rare condition might similarly be a special case for children, helping them resist their otherwise strong bias toward regularization. Investigating empirically how optionality interacts with frequency and other input characteristics in both children and adults could offer valuable insights into developmental differences in language acquisition.

5. Conclusion

In this paper, we proposed that optionality may be a particularly stable type of variation in natural language, and set out to demonstrate this observation across two artificial language learning experiments. While our results suggest that optionality is indeed a special type of variation, it was not in the ways we originally expected. Contrary to our initial hypothesis, adults strongly regularized when an optional form was frequent, suggesting optionality alone does not inherently resist the cognitive pressure to regularize. Further, to our surprise, adults exposed to a language with a frequent optional form regularized just as strongly as children have been shown to, within a single learning session. Our findings suggest that the cognitive mechanisms responsible for regularization likely extend beyond memory limitations, highlighting the need to consider other factors, such as how learners structure linguistic input or interpret forms in the presence of other distributional cues. Understanding these mechanisms could offer new insights into the broader cognitive processes underlying regularization, clarifying how both children and adults make use of various sources of information when encountering linguistic variation.

Data Accessibility Statement

Data and code are available at

https://osf.io/4nqdm/?view_only=f9b6bc0e24a744188e4f4cb5420ce866 (link is anonymized for peer review)

Ethics and consent

(Removed for peer review). This study was approved by the Institutional Review Board at XXX, protocol #XXX

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